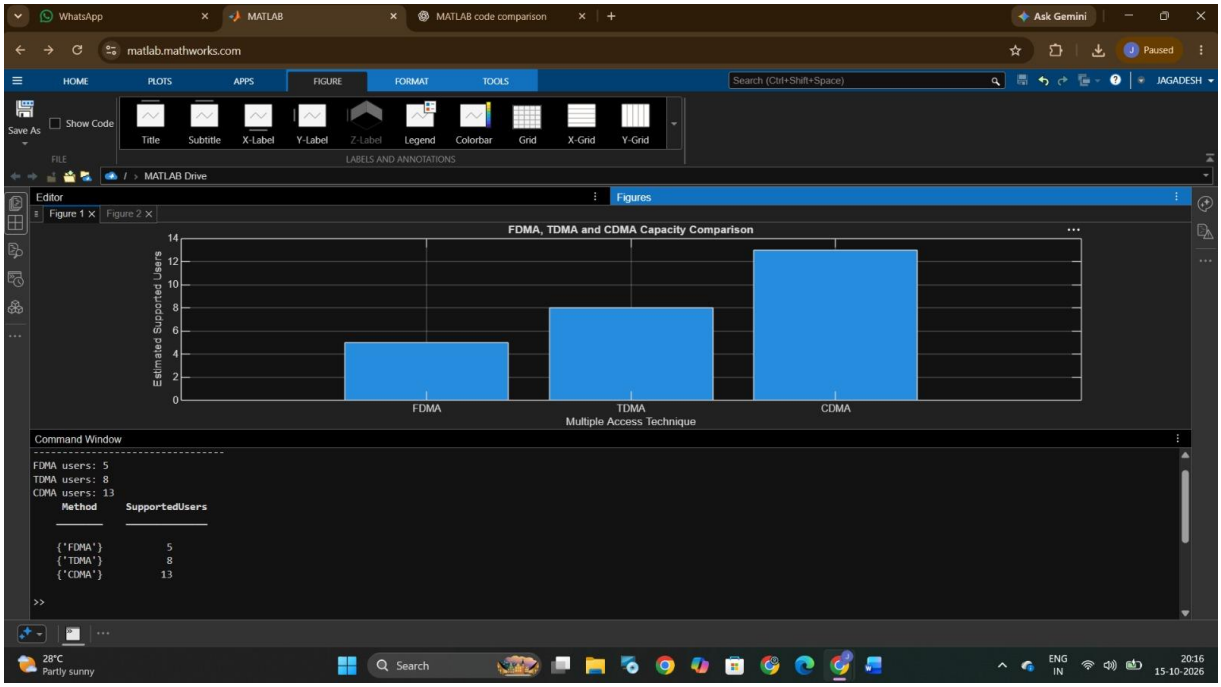
Experiment Number: 1 Simulation and Comparative Analysis of FDMA, TDMA and CDMA: Resource Allocation, User Capacity and Mobile-Generation Context

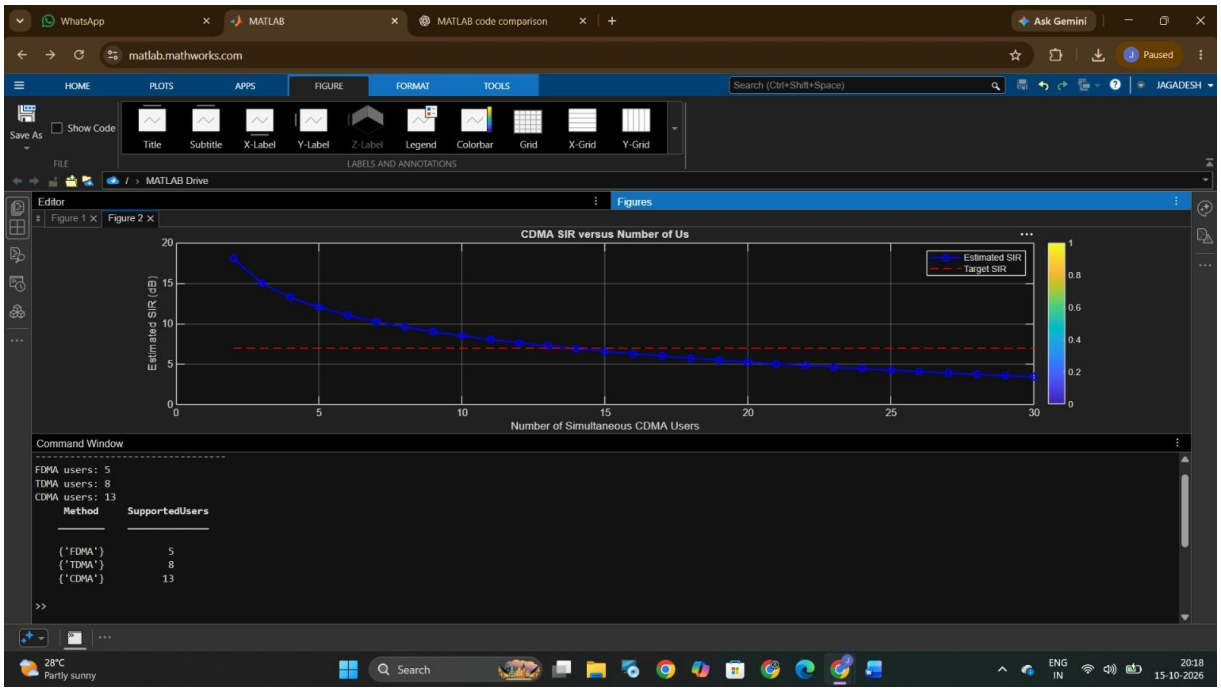
PROGRAM:

```
clear;
clc; close
all;
%% 1. Illustrative system parameters totalBandwidthHz = 1e6;
% Total bandwidth: 1 MHz fdmaChannelHz = 200e3;    %
FDMA channel width: 200 kHz tdmaSlots = 8;            % TDMA
slots per frame processingGain = 64;      % CDMA processing
gain targetSIRdB = 7;          % Minimum acceptable CDMA SIR
in dB
%% 2. Estimate supported users fdmaUsers =
floor(totalBandwidthHz / fdmaChannelHz); tdmaUsers =
tdmaSlots; targetSIR = 10^(targetSIRdB / 10); %
Simplified CDMA capacity estimate cdmaUsers = floor(1
+ processingGain / targetSIR); fprintf('Illustrative
maximum supported users\n'); fprintf('FDMA: %d users\n',
fdmaUsers); fprintf('TDMA: %d users\n', tdmaUsers);
fprintf('CDMA: %d users at a target SIR of %.1f dB\n\n', ...
    cdmaUsers, targetSIRdB); %% 3. Display
comparison table method = ["FDMA"; "TDMA";
"CDMA"]; supportedUsers = [fdmaUsers; tdmaUsers;
cdmaUsers]; results = table(method, supportedUsers);
disp(results);
%% 4. Plot estimated user capacity figure;
bar(categorical(method), supportedUsers);
ylabel('Illustrative maximum supported simultaneous users');
xlabel('Multiple access technique'); title('FDMA, TDMA and
CDMA: Capacity Estimate'); grid on;
%% 5. Estimate CDMA SIR as users increase
K = 2:30; sirLinear = processingGain ./ (K -
1); sirDb = 10 * log10(sirLinear); figure;
plot(K, sirDb, 'o-', 'LineWidth', 1.5); hold on;

yline(targetSIRdB, '--r', 'Target SIR'); xlabel('Number
of simultaneous CDMA users (K)'); ylabel('Estimated
SIR (dB)'); title('Estimated CDMA SIR versus
Number of Users'); grid on; hold off;
```

OUTPUT:


FDMA, TDMA and CDMA Capacity Comparison
Estimated Supported Users
FDMA
TDMA
CDMA
Multiple Access Technique
Command Window
FDMA users: 5
TDMA users: 8
CDMA users: 13
Method SupportedUsers
{'FDMA'} 5
{'TDMA'} 8
{'CDMA'} 13

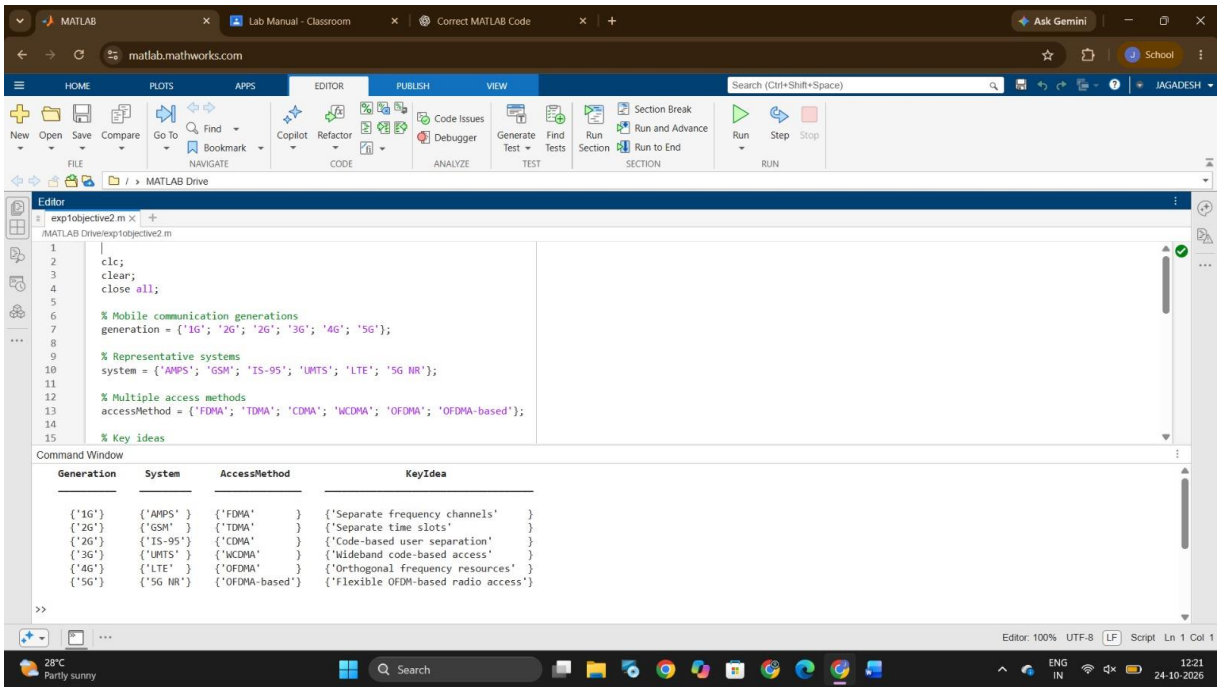

CDMA SIR versus Number of Us
Estimated SIR (dB)
Number of Simultaneous CDMA Users
Estimated SIR
Target SIR
Command Window
FDMA users: 5
TDMA users: 8
CDMA users: 13
Method SupportedUsers
{'FDMA'} 5
{'TDMA'} 8
{'CDMA'} 13

PROGRAM:

```
clear;
clc; close
all;
% Mobile communication generations generation =
["1G"; "2G"; "2G"; "3G"; "4G"; "5G"];
% Representative systems system = ["AMPS"; "GSM"; "IS-95";
"UMTS"; "LTE"; "5G NR"];
% Multiple access methods accessMethod = ["FDMA";
"TDMA"; "CDMA"; "WCDMA"; ...
            "OFDMA"; "OFDMA-based"]; %
Key idea of each system keyIdea = ["Analog
voice communication"; ...
        "Digital voice using time slots"; ...
        "Digital voice using spread-spectrum codes"; ...
        "Wideband code-division access"; ...
        "Orthogonal frequency-division access"; ...
        "Flexible OFDM-based radio access"];
% Create comparison table evolutionTable
= table( ...
   generation, system, accessMethod, keyIdea, ...
   'VariableNames', ...
   {'Generation', 'System', 'AccessMethod', 'KeyIdea'});
```

% Display table disp('Evolution of Mobile

Communication Systems'); disp(evolutionTable);



PROGRAM:

```
clear;
clc; close
all;
%% Create one figure with three diagrams figure('Color',
'w', 'Name', 'Multiple Access Techniques');

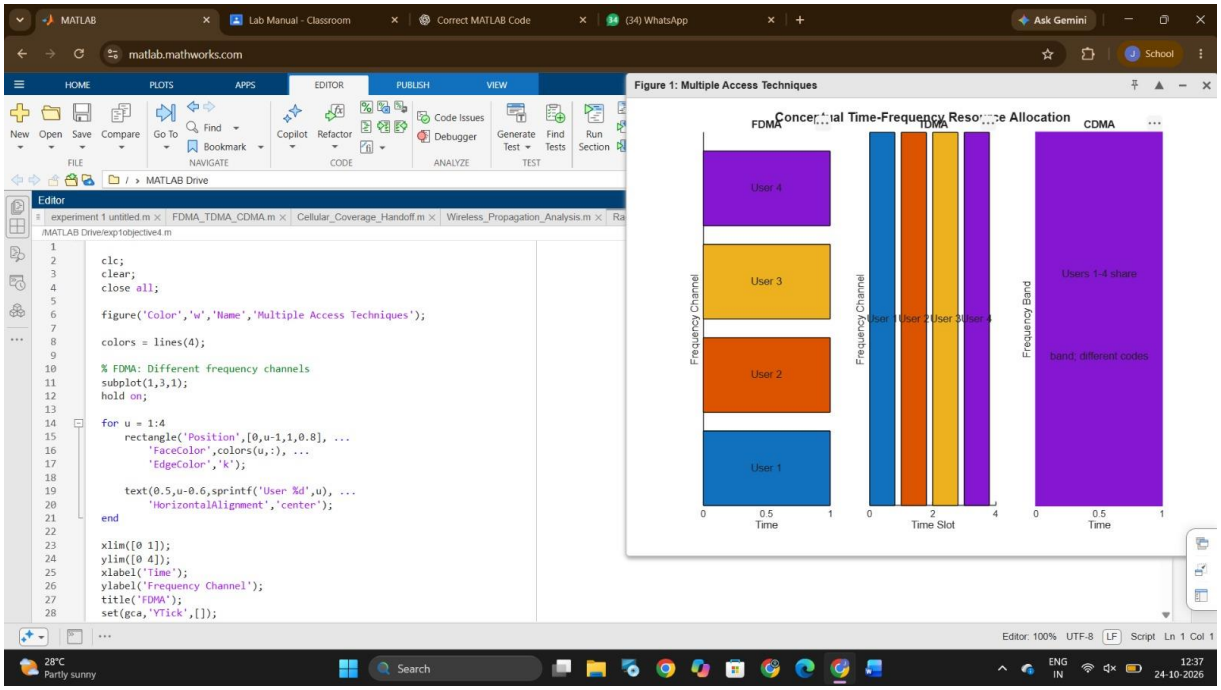
tiledlayout(1, 3, 'TileSpacing', 'compact', ...
             'Padding', 'compact'); colors
= lines(4);
%% 1. FDMA: Different frequency channels
nexttile; hold on; for u = 1:4
rectangle('Position', [0, u-1, 1, 0.8], ...
      'FaceColor', colors(u,:), ...
      'EdgeColor', 'k');    text(0.5, u-0.6,
sprintf('User %d', u), ...
'HorizontalAlignment', 'center'); end
xlim([0 1]); ylim([0 4]); xlabel('Time');
ylabel('Frequency channel');
title('FDMA'); set(gca, 'YTick', []);
```

```
box on; hold
off;
%% 2. TDMA: Different time slots
nexttile; hold on;

for u = 1:4     rectangle('Position', [u-1,
0, 0.8, 1], ...
       'FaceColor', colors(u,:), ...
'EdgeColor', 'k');     text(u-0.6, 0.5,
sprintf('User %d', u), ...
'HorizontalAlignment', 'center'); end
xlim([0 4]); ylim([0 1]); xlabel('Time
slot'); ylabel('Frequency channel');
title('TDMA'); set(gca, 'YTick', []); box
on; hold off;
%% 3. CDMA: Shared time-frequency resources
nexttile; hold
on;
% Illustrative overlapping user transmissions
for u = 1:4     rectangle('Position', [0, 0, 1, 1],
...
       'FaceColor', colors(u,:), ...
       'FaceAlpha', 0.15, ...
       'EdgeColor', colors(u,:), ...
       'LineWidth', 1.5); end
text(0.5, 0.62, 'Users 1-4 share', ...
'HorizontalAlignment', 'center');

text(0.5, 0.40, 'band; different codes', ...
'HorizontalAlignment', 'center');
xlim([0 1]); ylim([0 1]); xlabel('Time');
ylabel('Frequency band');
title('CDMA'); set(gca, 'YTick', []);
box on; hold off;
%% Overall title
sgtitle('Conceptual Time-Frequency Resource Allocation');
```

OUTPUT:


MATLAB
Lab Manual - Classroom
Correct MATLAB Code
(34) WhatsApp
Ask Gemini
matlab.mathworks.com
School
HOME
PLOTS
APPS
EDITOR
PUBLISH
VIEW
New
Open
Save
Compare
Go To
Find
Bookmark
Copilot
Refactor
Code Issues
Debugger
Generate Test
Find Tests
Run Section
FILE
NAVIGATE
CODE
ANALYZE
TEST
MATLAB Drive
Editor
experiment 1 untitled.m
FDMA_TDMA_CDMA.m
Cellular_Coverage_Handoff.m
Wireless_Propagation_Analysis.m
/MATLAB Drive/exp1objective4.m
clc;
clear;
close all;
figure('Color','w','Name','Multiple Access Techniques');
colors = lines(4);
% FDMA: Different frequency channels
subplot(1,3,1);
hold on;
for u = 1:4
rectangle('Position',[0,u-1,1,0.8], ...
'FaceColor',colors(u,:), ...
'EdgeColor','k');
text(0.5,u-0.6,sprintf('User %d',u), ...
'HorizontalAlignment','center');
end
xlim([0 1]);
ylim([0 4]);
xlabel('Time');
ylabel('Frequency Channel');
title('FDMA');
set(gca,'YTick',[]);
Figure 1: Multiple Access Techniques
FDMA
TDMA
CDMA
User 4
User 3
User 2
User 1
Frequency Channel
Time
Time Slot
Frequency Band
Users 1-4 share
band; different codes
Editor: 100%
UTF-8
Script
Ln 1 Col 1
28°C
Partly sunny
Search
ENG
IN
12:37
24-10-2026